> $f(b,c,r) := \sin^2(b) + \sin^2(c) + 2\cdot\sin(b)\cdot\sin(c)\cdot\cos(b+c+r)$

$$f := (b,c,r) \mapsto \sin(b)^2 + \sin(c)^2 + 2\cdot\sin(b)\cdot\sin(c)\cdot\cos(b+c+r) \quad \textbf{(1)}$$

> $g(b,c,r) := simplify\big(expand\big(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\big)\big)$

$$g := (b,c,r) \mapsto simplify\big(expand\big(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\big)\big) \quad \textbf{(2)}$$

> $collect(expand(g(b,c,r)), \cos(r))$

$$-4\sin(b)^2\sin(c)^2\cos(r)^2 + \big(-4\sin(b)^2\sin(c)^2\cos(c)^2 - 4\sin(b)^2\sin(c)^2\cos(b)^2 + 4\sin(b)\sin(c)\cos(c)^3\cos(b) + 4\sin(b)\sin(c)\cos(c)\cos(b)^3 + 8\sin(b)^2\sin(c)^2 - 8\sin(b)\sin(c)\cos(b)\cos(c)\big)\cos(r) - 4\sin(c)^4\sin(b)^2 + 4\sin(b)^2\sin(c)^2\cos(b)^2 - 4\sin(b)\sin(c)\cos(c)^3\cos(b) - 4\sin(b)\sin(c)\cos(c)\cos(b)^3 + 8\sin(b)\sin(c)\cos(b)\cos(c) \quad \textbf{(3)}$$

> $simplify\Bigg(\frac{1}{r^2}\Big(-4\sin(b)^2\sin(c)^2\cdot(1-r^2) + \big(-4\sin(b)^2\sin(c)^2\cos(c)^2 - 4\sin(b)^2\sin(c)^2\cos(b)^2 + 4\sin(b)\sin(c)\cos(c)^3\cos(b) + 4\sin(b)\sin(c)\cos(c)\cos(b)^3 + 8\sin(b)^2\sin(c)^2 - 8\sin(b)\sin(c)\cos(b)\cos(c)\big)\cdot\Big(1 - \frac{r^2}{2}\Big) - 4\sin(c)^4\sin(b)^2 + 4\sin(b)^2\sin(c)^2\cos(b)^2 - 4\sin(b)\sin(c)\cos(c)^3\cos(b) - 4\sin(b)\sin(c)\cos(c)\cos(b)^3 + 8\sin(b)\sin(c)\cos(b)\cos(c)\Big)\Bigg)$

$$-2\,\big(\cos(c)\cos(b)^3 - \sin(b)\sin(c)\cos(b)^2 + \big(\cos(c)^3 - 2\cos(c)\big)\cos(b) - \sin(b)\sin(c)\cos(c)^2\big)\sin(b)\sin(c) \quad \textbf{(4)}$$

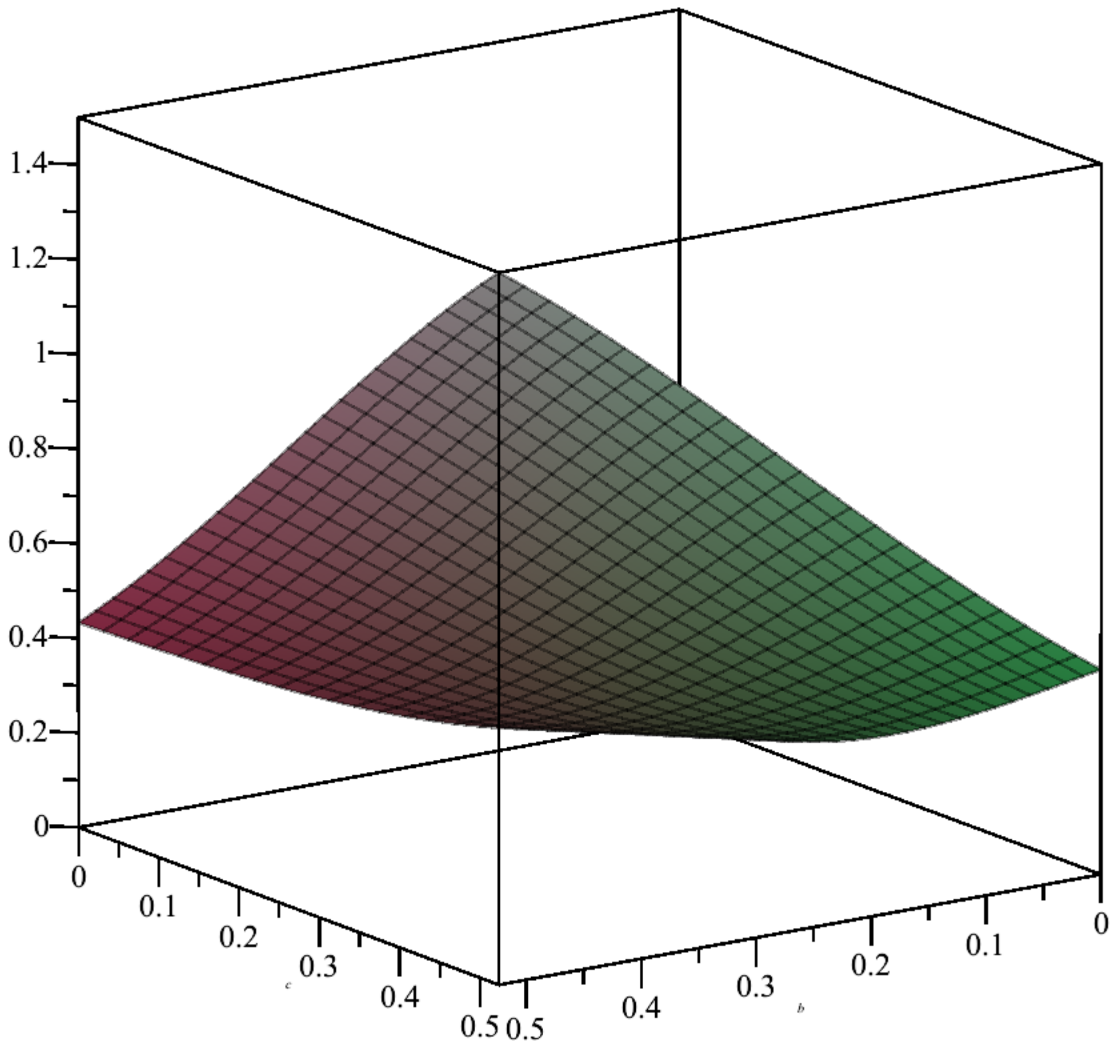
> $h(b,c) := -2\,\big(\cos(c)\cos(b)^3 - \sin(b)\sin(c)\cos(b)^2 + \big(\cos(c)^3 - 2\cos(c)\big)\cos(b) - \sin(b)\sin(c)\cos(c)^2\big)\sin(b)\sin(c)$

$$h := (b,c) \mapsto -\big(2\cdot\cos(c)\cdot\cos(b)^3 - 2\cdot\sin(b)\cdot\sin(c)\cdot\cos(b)^2 + 2\cdot\big(\cos(c)^3 - 2\cdot\cos(c)\big)\cdot\cos(b) - 2\cdot\sin(b)\cdot\sin(c)\cdot\cos(c)^2\big)\cdot\sin(b)\cdot\sin(c) \quad \textbf{(5)}$$

> $simplify\big(h(b,c) - 2\cdot\big(\sin^2(b) + \sin^2(c)\big)\cdot\cos(b+c)\cdot\sin(b)\sin(c)\big)$

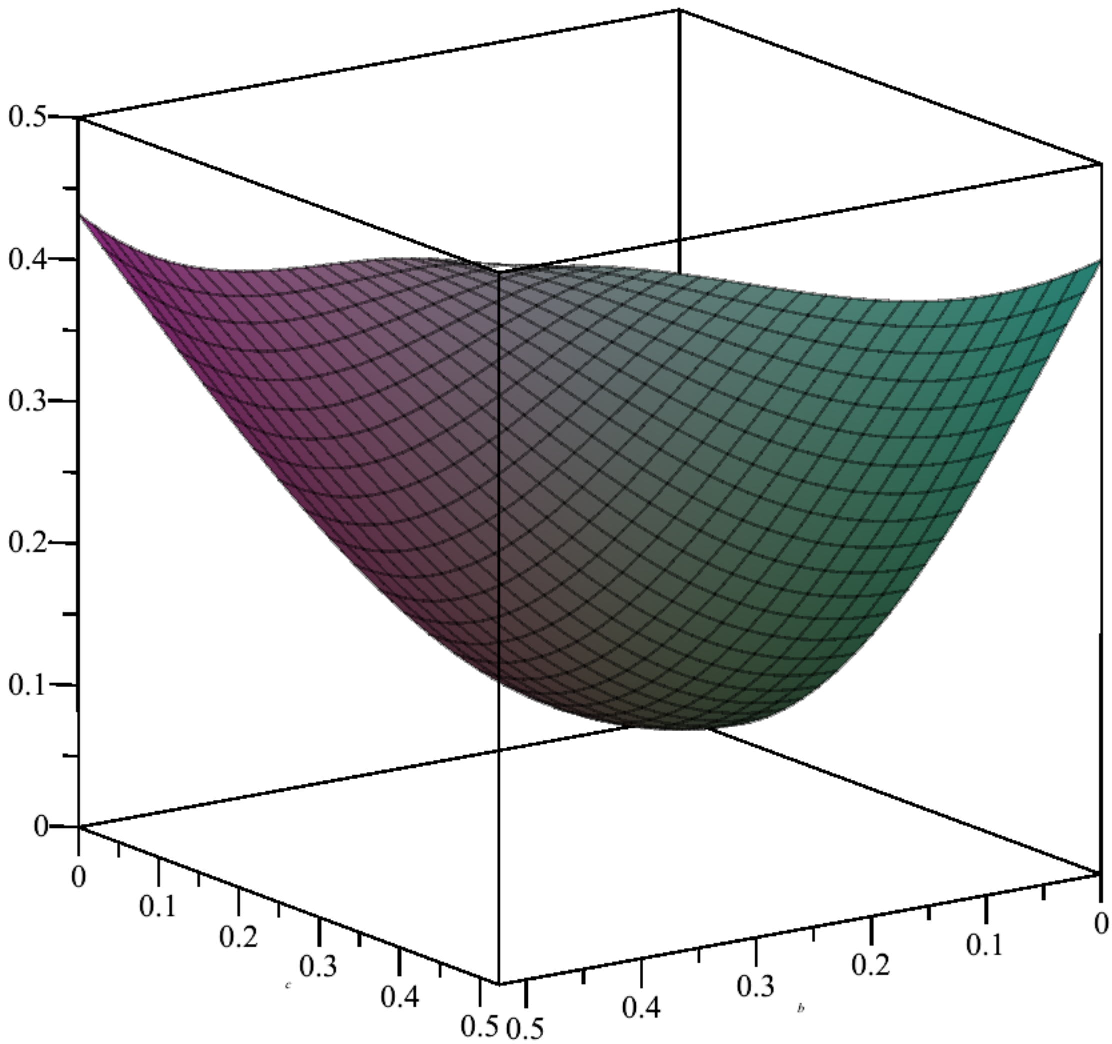
$$4\sin(b)^2\sin(c)^2 \quad \textbf{(6)}$$

> $plot3d\left(\frac{h(b,c)}{\sin(b)\sin(c)}, b = 0..\frac{3.14}{6}, c = 0..\frac{3.14}{6}\right)$



1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
c
0.5
0.4
0.3
b
0.2
0.1
0

>



$$> \quad int\left(int\left(\frac{h(b,c)}{f(c,b,0)^2}, b = 0..\frac{3.14159265359}{6}, numeric = true\right), c = 0..\frac{3.14159265359}{6}, numeric = true\right)$$

$$0.1140514567 \tag{7}$$

$$> \quad \exp\left(\frac{\pi^2}{8} - 3\cdot 0.1140514567 - \frac{3}{2}\cdot 0.1438410363 - 3\cdot 0.1366658305\right)$$

$$1.304457355 \tag{8}$$

>